

The Effect of Public Wifi Connections on User Security

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Part 2 - Engineering Problem and Goal

- The majority of people accessing public wi-fi networks do not use a VPN to encrypt their data, however, it's unclear whether this is of concern because the use of HTTP (unencrypted connections) is practically obsolete. Is the inconvenience of connecting to a VPN and the possibility of data breaches when using free VPN providers necessary if browsing is restricted to HTTPS (encrypted) sites?
- In previous studies, researchers explored the amount of people who accessed these sites publicly, and were able to find some insecure data.
 - “The percentage of users who access the encrypted sites is much higher than unencrypted one[s], but we also found some potential risks should be of concern when accessing information in public wifi places.” (Batam Case 2019)
- To expand on this, the additional factor of VPN usage is considered, along with the accessibility of data when using HTTPS sites.
- Goal: Find out whether VPN use is necessary while accessing basic data on HTTPS websites.

Part 3 - Procedures

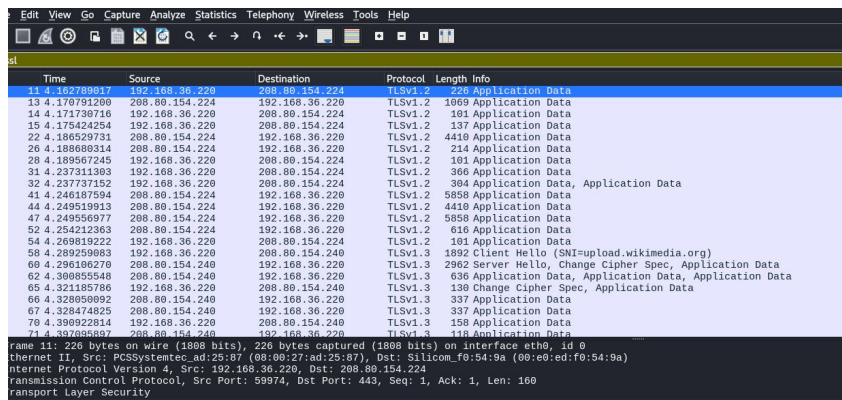
- Installed a Kali Linux Virtual Machine on laptop, to access Wireshark in order to collect network packets
- Captured data from 16 different websites → 8 HTTP, and 8 HTTPS, first without, then while connected to a VPN. Afterwards, I conducted additional trials at 2 other public wi-fi locations. Used capture filters in order to avoid capturing data from other users.
- Analyzed data in Wireshark, searching for instances of insecure data, and tracing patterns based on location, HTTP/HTTPS being accessed, and presence of VPN
 - See figure 1 for HTTPS instance of generally secure data.
 - See figure 2 & 3 for pattern demonstrating difference between data collected with vs without a VPN.
 - See figure 4 for HTTP instance of insecure data.

The image shows a Wireshark packet capture window. The top pane displays a list of network packets. Packet 760 is selected, showing a TLSv1.3 Client Hello packet. The bottom pane shows the details of this packet, including the TLSv1.3 Client Hello structure with fields like EC point formats, Elliptic curves point formats, and Extensions. The packet is encrypted, and the details pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
744	28.388317354	44.215.116.52	192.168.1.153	TCP	66	443 → 58170 [ACK] Seq=
745	28.315978695	44.215.116.52	192.168.1.153	TCP	66	443 → 58170 [ACK] Seq=
746	28.315979023	44.215.116.52	192.168.1.153	TCP	66	443 → 58170 [ACK] Seq=
747	28.324443256	44.215.116.52	192.168.1.153	TCP	66	443 → 58170 [ACK] Seq=
748	28.324443641	44.215.116.52	192.168.1.153	TCP	66	443 → 58170 [ACK] Seq=
749	28.337927705	44.215.116.52	192.168.1.153	TLSv1.3	682	Application Data
750	28.337975310	192.168.1.153	44.215.116.52	TCP	66	58170 → 443 [ACK] Seq=
751	29.246198263	192.168.1.153	192.168.1.1	DNS	89	Standard query 0xfdbf
752	29.246528491	192.168.1.153	192.168.1.1	DNS	89	Standard query 0x833d
753	29.246729960	192.168.1.153	192.168.1.1	DNS	89	Standard query 0x1807
754	29.257893669	192.168.1.1	192.168.1.153	DNS	170	Standard query respons
755	29.257893891	192.168.1.1	192.168.1.153	DNS	170	Standard query respons
756	29.259800783	192.168.1.1	192.168.1.153	DNS	153	Standard query respons
757	29.301743180	192.168.1.153	3.162.130.8	TCP	74	38322 → 443 [SYN] Seq=
758	29.310544176	3.162.130.8	192.168.1.153	TCP	74	443 → 38322 [SYN, ACK]
759	29.310572822	192.168.1.153	3.162.130.8	TCP	66	38322 → 443 [ACK] Seq=
760	29.310843322	192.168.1.153	3.162.130.8	TLSv1.3	1637	Client Hello (SN=039x
761	29.320017719	3.162.130.8	192.168.1.153	TCP	66	443 → 38322 [ACK] Seq=
762	29.320017758	3.162.130.8	192.168.1.153	TLSv1.3	4350	Server Hello, Change C
763	29.320039030	192.168.1.153	3.162.130.8	TCP	66	38322 → 443 [ACK] Seq=

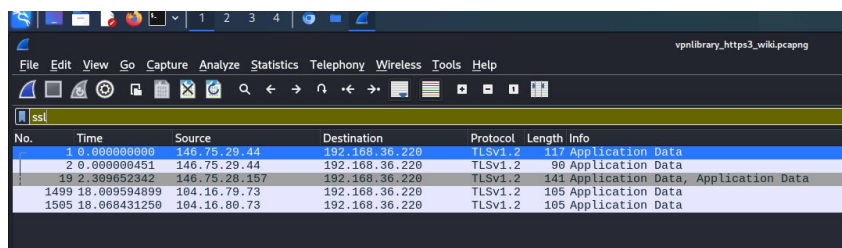
Figure 1: Capture of home_https4_amazon, a HTTPS site, consisting of TLS (HTTPS) packets. All data inside is encrypted, any plaintext is not compromising to security threats/hackers. However, DNS packets are presented, which allow internet service providers to partially see your data (not at risk for normal civilian attacks).

Part 3 - Procedures



No.	Time	Source	Destination	Protocol	Length	Info
11	4.102789017	192.168.36.220	208.80.154.224	TLSv1.2	226	Application Data
13	4.170191200	208.80.154.224	192.168.36.220	TLSv1.2	1059	Application Data
14	4.171738716	192.168.36.220	208.80.154.224	TLSv1.2	101	Application Data
15	4.175424254	192.168.36.220	208.80.154.224	TLSv1.2	137	Application Data
22	4.186529731	208.80.154.224	192.168.36.220	TLSv1.2	4419	Application Data
26	4.188680314	208.80.154.224	192.168.36.220	TLSv1.2	214	Application Data
28	4.189567245	192.168.36.220	208.80.154.224	TLSv1.2	101	Application Data
31	4.237311383	192.168.36.220	208.80.154.224	TLSv1.2	366	Application Data
32	4.237737152	192.168.36.220	208.80.154.224	TLSv1.2	384	Application Data, Application Data
41	4.246187594	208.80.154.224	192.168.36.220	TLSv1.2	5858	Application Data
44	4.249519913	208.80.154.224	192.168.36.220	TLSv1.2	4419	Application Data
47	4.249556977	208.80.154.224	192.168.36.220	TLSv1.2	5858	Application Data
52	4.254212363	208.80.154.224	192.168.36.220	TLSv1.2	616	Application Data
54	4.268619222	192.168.36.220	208.80.154.224	TLSv1.2	101	Application Data
58	4.289259083	192.168.36.220	208.80.154.240	TLSv1.3	1892	Client Hello (SNI=upload.wikimedia.org)
60	4.296186270	208.80.154.240	192.168.36.220	TLSv1.3	2962	Server Hello, Change Cipher Spec, Application Data
62	4.308855548	208.80.154.240	192.168.36.220	TLSv1.3	606	Application Data, Application Data, Application Data
65	4.321185786	192.168.36.220	208.80.154.240	TLSv1.3	130	Change Cipher Spec, Application Data
66	4.328050892	208.80.154.240	192.168.36.220	TLSv1.3	337	Application Data
67	4.328474825	208.80.154.240	192.168.36.220	TLSv1.3	337	Application Data
70	4.390922814	192.168.36.220	208.80.154.240	TLSv1.3	150	Application Data
71	4.392695887	208.80.154.240	192.168.36.220	TLSv1.3	118	Application Data

Figure 2

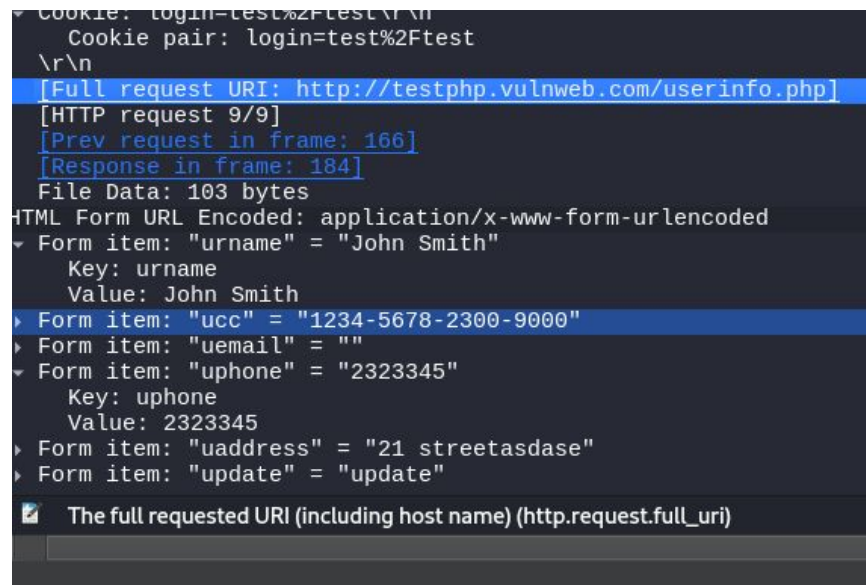


No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	146.75.29.44	192.168.36.220	TLSv1.2	117	Application Data
2	0.000000451	146.75.29.44	192.168.36.220	TLSv1.2	90	Application Data
19	2.309652342	146.75.28.157	192.168.36.220	TLSv1.2	141	Application Data, Application Data
1499	18.009594899	104.16.79.73	192.168.36.220	TLSv1.2	105	Application Data
1505	18.068431258	104.16.80.73	192.168.36.220	TLSv1.2	105	Application Data

Figure 3

Figures 2 & 3: Comparison of library_https3_wiki (Figure 2) and vpnlibrary_https3_wiki (Figure 3). It can be seen that significantly more packets were collected in the non-vpn trial. Demonstrates that VPN prevents most data from being traced on the network, while HTTPS data is seen, but isn't useful for hackers since it is encrypted

- Data collected demonstrates my goal of comparing different captures, specifically HTTP vs HTTPS, and VPN vs non-VPN.
- Analyzed data by putting in filters and searching for instances + the possibility of insecure data.



```
Cookie: login=test%2Ftest\r\n
Cookie pair: login=test%2Ftest
\r\n
[Full request URI: http://testphp.vulnweb.com/userinfo.php]
[HTTP request 9/9]
[Prev request in frame: 166]
[Response in frame: 184]
File Data: 103 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "urname" = "John Smith"
Key: urname
Value: John Smith
Form item: "ucc" = "1234-5678-2300-9000"
Form item: "uemail" = ""
Form item: "uphone" = "2323345"
Key: uphone
Value: 2323345
Form item: "uaddress" = "21 streetasdase"
Form item: "update" = "update"
The full requested URI (including host name) (http.request.full_uri)
```

Figure 4: Capture of library_http5_vulnweb, a HTTP site, exposing “sensitive” data simulated using fake login details. Confirms insecurity of http websites, as they store data in plaintext (not encrypted)

Part 4 - Results

Pros and Cons of using a VPN vs. not using one and only accessing secure HTTPS connections when on public wi-fi networks:

VPN

- Visit any site without network owner or websites seeing your IP address or DNS traffic (see Table 1)
- Inconvenient, slower connection
- Some Free VPN providers log your data and sell it, making it more insecure (I used ProtonVPN which reported that it does not log any data)
- Must research a trustworthy VPN provider, some charge money, many offer paid versions

No VPN + HTTPS:

- Only visit HTTPS Sites
- If Secure DNS settings are not enabled, the network owner can see what websites you are accessing
- Websites can see your IP address, may keep track of your data
- No one can see your login info/what you are searching up on HTTPS sites

Overall, my research met my goals of showing whether HTTPS security is enough to be used without a VPN, and regardless, that VPNs do provide an additional layer of security that is necessary when accessing highly-sensitive information. I was able to confirm this by looking at HTTPS packets in comparison to HTTP and VPN packets, and compiling my results into data tables in my Packet Tracking Log.

Part 4 - Results

Table 1: Data collected + pattern analysis when using a VPN. Found that no data collected was under TLS (HTTPS) or HTTP (indicated as 0% of both), packets were often only listed as UDP, meaning that only data from the VPN connection (which uses UDP traffic) was traceable on the network. In some instances, “No Packets collected,” when VPN server prevented logging its own data (not even UDP packets appeared during packet collection)

Trial	Home	Library	Paris Baguette
https1_google	0% of both	0% of both	0% of both
https2_yt	0% of both	No Packets collected	0% of both
https3_wiki	0% of both	No Packets collected	0% of both
https4_amazon	0% of both	TLS - 0% 1 Packet collected	0% of both
https5_drive	0% of both	TLS - 0.3%	0% of both
https6_rrhs	0% of both	No Packets collected	0% of both
https7_apple	0% of both	TLS - 0% 4 Packet collected	0% of both
https8_bing	0% of both	TLS - 0% 1 Packet collected HTTP - 0.1%	0% of both
http1_eprints	0% of both	No Packets collected	0% of both
http2_woodwind	0% of both	No Packets collected	0% of both
http3_old	0% of both	No Packets collected	0% of both
http4_text	0% of both	No Packets collected	0% of both

Table 2: Data collected + pattern analysis while not using a VPN. Amount of TLS + HTTP data varies based off of website, but also seems to be impacted by location. Also, most HTTPS packets only captured TLS data and not HTTP data, implying that it is more secure. Insecurity can surface with unencrypted HTTP cookies, with small amounts of HTTP packets collected while using HTTPS websites.

https6_rrhs	0% of both	TLS - 7.6%	TLS - 3.9%
https7_apple	TLS - 99.1%	TLS - 43.7%	TLS - 45.4%
https8_bing	TLS - 91.2%	TLS - 11.8%	TLS - 14%
http1_eprints	TLS - 0.1% HTTP - 37.1%	TLS - 15.1% HTTP - 2.6%	TLS - 0.2% HTTP - 10.9%
http2_woodwind	TLS - 18.2% HTTP - 4.7%	TLS - 5.1% HTTP - 0.2%	TLS - 5.6% HTTP - 0.2%
http3_old	TLS - 12.9% HTTP - 4.4%	TLS - 11.6% HTTP - 0.6%	TLS - 9.5% HTTP - 1.2%

Part 5 - Discussion

The results of this experiment were able to fulfill my overall goal, to determine the necessity of a VPN if only accessing basic data on HTTPS websites, showing that HTTPS alone offered a high level of security, which aligned with my previous hypotheses for this study. The results also indicated that using a VPN completely prevents both TLS and HTTP packet collection, which aligns with commonly held beliefs and published data that state that VPN usage offers significant security. Overall, I did expect that the VPN results would provide more security due to previous research, but this experiment also showed that accessing HTTPS connections without a VPN isn't as dangerous as some make it out to be.

A few factors that may have influenced my results include that I didn't access same tabs/number of tabs on each website, accessed for different amounts of time, and that I didn't use the exact VPN connection every time, auto connected to different ones based on location. The tabs could have impacted the percentages of TLS/HTTP packets collected, because I could have been accessing the tabs for different amounts of time during each trial, and accessed different tabs each time. The VPN connection caused a discrepancy in collecting VPN packets, as in some trials the VPN's data collected came up as UDP traffic, and in other cases it didn't show up at all (See Table 1, instances of "No Packets Collected")

One issue that arose during the experiment was the significant amounts of data pulled and it was difficult to look for specific instances of insecurity. I addressed this using additional filters to scan through the data more efficiently.

My experiment is an advancement over current resources for public wi-fi safety because it focuses on the general user and whether they would be able to access their data without a VPN for basic needs, or if a free VPN is necessary using direct packet captures. Overall, it still urges VPN use when accessing highly sensitive data, but provides an alternative for day-to-day usage.

Part 6 - Conclusions

I met my goals of recording data when accessing the websites under different conditions, as I collected data at 3 different locations, both with and without a VPN. See Table 1 for reference of data collected while using a VPN.

Meaning of Results:

- This study's results promote using a VPN if you are able to get a paid VPN service, and/or if you are accessing sensitive data (i.e. bank statements, office software, personally-identifiable information), and especially if you plan to access HTTP websites, when on a public wi-fi network.
- However, if you can only access a free VPN service, you plan to only access trusted HTTPS websites, and you won't be accessing/inputting any sensitive data, it is secure for you to not use a VPN.
- If you are not using a VPN, turn on "Secure DNS" settings so that your internet service provider or public wi-fi network owner cannot see what websites you are accessing.

Potential Applications:

- Comparing the study results when using different VPN providers, especially in Free vs. Paid services
- Researching methods to see whether a Free VPN may be logging or selling data
- Accessing HTTPS sites while imitating attacks over the network to see the effectiveness
- Comparing data security when using Secure DNS vs. Default.

Part 7 - References

(PDF) network traffic analysis and Intrusion Detection Using Packet Sniffer. (n.d.-b).

https://www.researchgate.net/publication/232625696_Network_Traffic_Analysis_and_Intrusion_Detection_Using_Packet_Sniffer

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